



GETTING STARTED WITH VERSION CONTROL: INTRO TO GIT AND GITHUB



THE UNIVERSITY of EDINBURGH
Centre for Data, Culture & Society



Welcome!



Aislinn Keogh

PhD student in linguistics



www.cdcs.ed.ac.uk

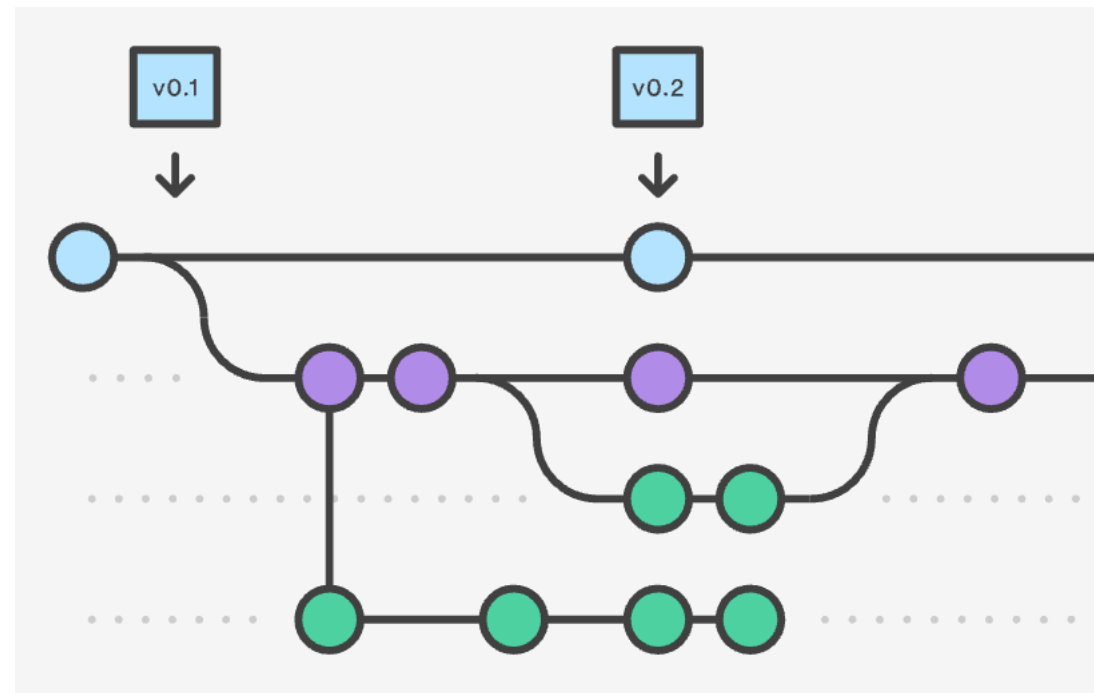
Plan for Today

- 10:30** Welcome & introduction
- 10:50** *Live demo & troubleshooting:* Create your first repo & your first commit
- 11:10** *Live Demo & troubleshooting:* Create a new branch and merge it
- 11:30** Break
- 11:40** *Live Demo & troubleshooting:* Fork a repo & create a pull request
- 12:00** *Live Demo & troubleshooting:* Resolving a merge conflict
- 12:20** Wrap up & additional resources
- 12:30** End of workshop



Version Control

- **Version control is the practice of tracking and managing *changes* to files (traditionally, software code).**
- Version control software keeps track of every modification to the files in a special kind of database.
- Version control tracks every individual change by each contributor and helps prevent *concurrent* work from conflicting.
- It is a standard *workflow* in the tech industry.



git

A distributed version control system

No need to learn this for basic use cases!

```
singh@DESKTOP-PGVSHMF MINGW64 ~
$ git checkout master
Switched to branch 'master'

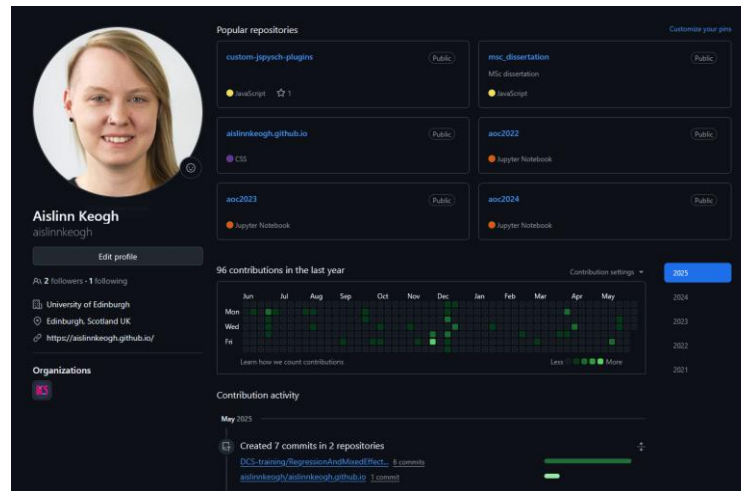
singh@DESKTOP-PGVSHMF MINGW64 ~
$ git merge newbranch
Updating 9e7f7d0..3eb93e9
Fast-forward
 branch file.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch file.txt

singh@DESKTOP-PGVSHMF MINGW64 ~/Desktop/newRepo (master)
$ |
```

Image from <https://funkyvast.weebly.com/what-is-git-bash-terminal.html>

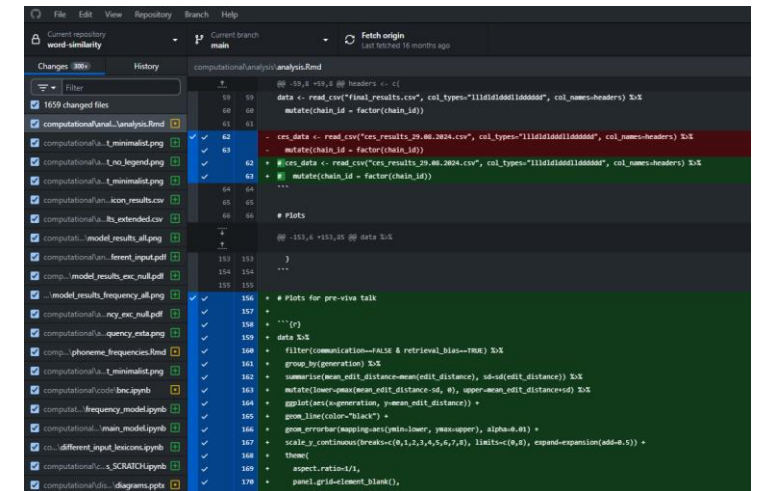
GitHub

An online platform for collaborating on code, based on git



GitHub Desktop

A desktop application for working with git and GitHub on your computer



Terminology

repository/repo

commit

clone

branch

merge

fetch

pull

push

fork



Confused Person - ClipArt B...
clipartbest.com



Confused | Free Images at Clker...
clker.com



Confused Look Emoticon | ww...
imgkid.com



Confused Man Fre...
publicdomainpictur...



http://www.dreamstime.com/royalty-...
happyhormonesforlife.com



Confuse gesture, Confusion Medical s...
pngegg.com



Image Of Confused Person...
cliparts.co



Confused Cartoon ...
cliparts.co



Textusa: Perhaps... Confusing
textusa.blogspot.com



Confused person png imag...
lovepik.com



January | 2010 | Investing Caffeine
investingcaffeine.com



Principals Share The Most Bizarre ...
humaverse.com



Don't Worry, We're All Confused | Show...
showmeinstitute.org



confused frustrated business man dreamst...
jobcrusher.com

Terminology

- repository/repo** → a “folder” for your code for one project
- commit** → saving your changes in the project
- clone** → a copy of a repo (e.g. on your computer)
- branch** → a parallel version of a repo (repos can have multiple branches)
- merge**
- fetch**
- pull**
- push**
- fork**



Live Demo:

Creating your first repo

1. Go to www.github.com
2. Sign in / sign up
3. Click the + in the top right corner and select “New repository”
4. Choose a repository name, check “Add a README file”, and leave all other fields.
5. Click “Create repository”



Live Demo:

Cloning your repo with GitHub Desktop

1. Install GitHub Desktop from <https://desktop.github.com> if you haven't yet
2. On the GitHub page for your repo, click “Open in GitHub Desktop”
3. Choose where to clone your repo
4. Click **fetch** & admire your repo in GitHub Desktop



Live Demo:

Making a commit

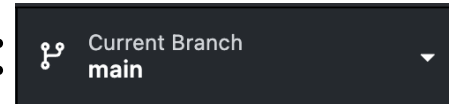
1. Open the README file in a text editor of your choice
If you're unsure: on Windows use **Notepad**; on Mac use **TextEdit**
2. Make some changes in your README file and save them
3. In GitHub Desktop, describe your commit and click commit when ready
4. Push to GitHub



Live Demo:

Creating a new branch and merging it

1. In Github Desktop, open the branch menu:
2. Select 'new branch' and enter a name of your choice. Confirm that this branch is now displayed under 'Current Branch'.
3. Make some changes in your README file and save them as before, commit, then push.
4. Switch back to the main branch, click 'Choose a branch to merge into main' in the branch dropdown, select your new branch, and merge.



The background of the slide is a collage. The top half features a black and white photograph of a classroom or office from the mid-20th century, with several people seated at desks. A map is visible on the wall. The bottom half features a stylized, high-contrast red and black illustration of a woman in a floral dress holding a book. The word 'Break' is centered in a large blue font across the middle of the collage.

Break

**GETTING STARTED WITH
VERSION CONTROL: INTRO
INTRO TO GIT AND GITHUB**

Terminology

- repository/repo** → a “folder” for your code for one project
- commit** → saving your changes in the project
- clone** → a copy of a repo (e.g. on your computer)
- branch** → a parallel version of a repo (repos can have multiple branches)
- merge** → combine the changes from one branch into another
- fetch** → getting the latest changes from the online repo
- pull** → integrating the latest changes into your clone of the repo
- push** → sending your changes (your commits) to the server
- fork** → a personal copy of someone else’s repo



Live Demo:

Forking a repo, making a commit, and making a pull request

1. Find a partner and get the link to their GitHub repo. Fork it on GitHub, then add your fork to GitHub Desktop.
2. Make a change to their README, commit, and push.
3. On GitHub, view your fork and open a pull request.
4. Review and approve each other's pull requests.



Live Demo:

Resolving a merge conflict

1. Choose either your partner's or your repo.
2. To create a merge conflict: At the same time, both of you edit the title of the README, commit, and push.
3. The person who pushed second will get an error message about a merge conflict: resolve this as demonstrated.



Additional Resources

- GitHub online tutorials
 - Introduction (tutorial repeating some of what we did today): <https://github.com/skills/introduction-to-github>
 - Reviewing pull requests: <https://github.com/skills/review-pull-requests>
 - Resolving merge conflicts: <https://github.com/skills/resolve-merge-conflicts>
 - List of all skills tutorials: <https://github.com/skills>
- GitHub student pack: <https://education.github.com/pack>

